

INTERMEDIATE LEVEL • 2025 EDITION

Intermediate AI Engineering

From your first LangChain app to full-stack AI products.
Hands-on projects that build real skills.

8

Episodes

~4h

Total Runtime

5+

Projects Built

100%

Code Included

By **Rohith** • AI Engineer & Educator

Course Episodes

8 episodes taking you from LangChain basics to deploying full-stack AI applications

EP 1

THE COURSE

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outputs into exactly what you need — every single time.

EP 1

Building Your First AI App with LangChain

Go from zero to a working AI application using LangChain, OpenAI, and Python in one episode.

- ✓ LangChain architecture and core concepts
- ✓ Chains, prompts, and output parsers
- ✓ Building a Q&A chatbot from scratch
- ✓ Memory management for conversations
- ✓ Deploying your first AI app

EP 2



EP 2

Vector Databases & Semantic Search

Learn how AI finds information — build a semantic search engine that understands meaning, not just keywords.

- ✓ How embeddings represent meaning
- ✓ Pinecone, Weaviate, and ChromaDB compared
- ✓ Building a semantic search engine
- ✓ Similarity search and nearest neighbors
- ✓ Real-world use cases and performance tips

EP 3



EP 3

RAG: Retrieval Augmented Generation

Connect your AI to your own data. Build a document Q&A system that actually knows your content.

- ✓ RAG architecture explained visually
- ✓ Document ingestion and chunking strategies
- ✓ Vector retrieval and context injection
- ✓ Handling long documents
- ✓ Evaluation: is your RAG working?

**EP 4****EP 4**

Building AI Agents with Tools

Your AI can now take actions — search the web, run code, call APIs. Build your first autonomous agent.

- ✓ What makes an agent vs a chatbot
- ✓ Tool definition and function calling
- ✓ ReAct reasoning and action loop
- ✓ Web search, code execution, API tools
- ✓ Error handling and agent safety

**EP 5****EP 5**

Prompt Engineering Mastery

Get consistently better results from any LLM with proven prompting techniques used by AI engineers.

- ✓ Chain-of-thought and few-shot prompting
- ✓ Role prompting and persona techniques
- ✓ Structured output extraction
- ✓ Prompt templates and versioning
- ✓ Measuring prompt quality objectively

**EP 6****EP 6**

AI APIs: OpenAI, Anthropic & Beyond

Master the major AI APIs — understand the differences, costs, and when to use which model.

- ✓ OpenAI GPT-4 API deep dive
- ✓ Anthropic Claude API patterns
- ✓ Function calling and structured outputs
- ✓ Token costs and optimization
- ✓ Switching between providers

**EP 7****EP 7**

Streaming, Async & Production Patterns

Ship AI apps that feel fast. Learn streaming responses, async patterns, and production best practices.

- ✓ Streaming with Server-Sent Events
- ✓ Async LLM calls for throughput
- ✓ Rate limiting and retry logic
- ✓ Caching for cost reduction
- ✓ Logging and observability basics

A man with dark hair and a beard, wearing a black sweater, is speaking into a professional microphone. He is positioned in front of a bookshelf filled with books and decorated with string lights. The scene is dimly lit, focusing on the speaker.

EP 8

EP 8

Building a Full-Stack AI App

Put it all together — build a complete full-stack AI application with a beautiful UI and real backend.

- ✓ FastAPI backend with LangChain
 - ✓ React frontend with streaming UI
 - ✓ Authentication and user management
 - ✓ Deployment to Railway and Vercel
 - ✓ Monitoring and error tracking
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Ready for the Next Level?

After completing this course, continue with the Advanced AI Engineering course to master production systems, fine-tuning, and agent frameworks.

Advanced Course: <https://youtu.be/VX29jXVsyAI>